

B2 - BACnet / Modbus (Advanced) Modbus - Workshop I

6-way EPIV4

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6-way EPIV4 - Network

Modbus Poll



EP015R6+BAC-HHM

— Serial network (RS485)

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1. Power your device.

 *See wiring diagrams in the datasheet*

2. Connect your device via Modbus RTU with your laptop.

 *Use RS485-USB adapter*

 *See wiring diagram in the datasheet*

3. Configure your actuator according to network overview.

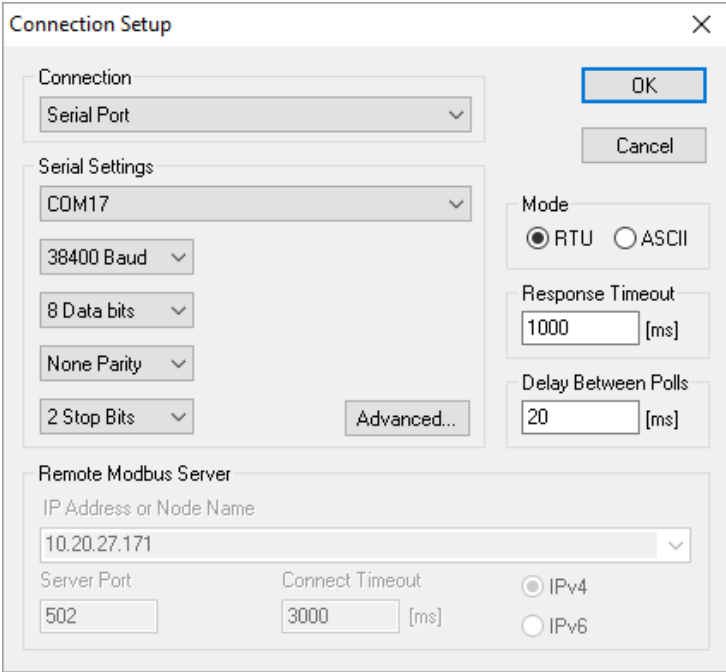
 *Find your appropriate configuration tool in the documents (LINK.10, Belimo Assistant 2, etc.)*

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4. Start Modbus Poll. Open a *Serial Port Connection*.

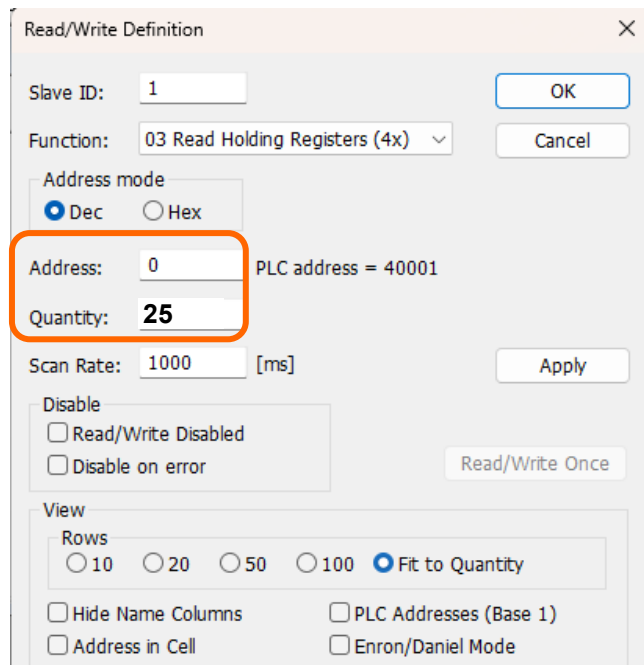
- 💡 Check on which COM port your RS485-USB adapter is
- 💡 Pay attention to the Modbus settings on your actuator you just made



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5. Open a new window  and press the F8 key.
 Make the settings according to the picture on the left.
 Repeat the procedure with the settings as shown in the picture on the right.



Read/Write Definition

Slave ID: 1

Function: 03 Read Holding Registers (4x)

Address mode: ☒ Dec ☐ Hex

Address: 0 PLC address = 40001

Quantity: 25

Scan Rate: 1000 [ms]

Disable: ☐ Read/Write Disabled ☐ Disable on error

View: Rows ☐ 10 ☐ 20 ☐ 50 ☐ 100 ☒ Fit to Quantity

☐ Hide Name Columns ☐ PLC Addresses (Base 1)

☐ Address in Cell ☐ Enron/Daniel Mode

OK Cancel Apply Read/Write Once

Address: 53

Quantity: 12

Address: 99

Quantity: 22

Address: 147

Quantity: 3

Address: 157

Quantity: 2

Address: 200

Quantity: 20

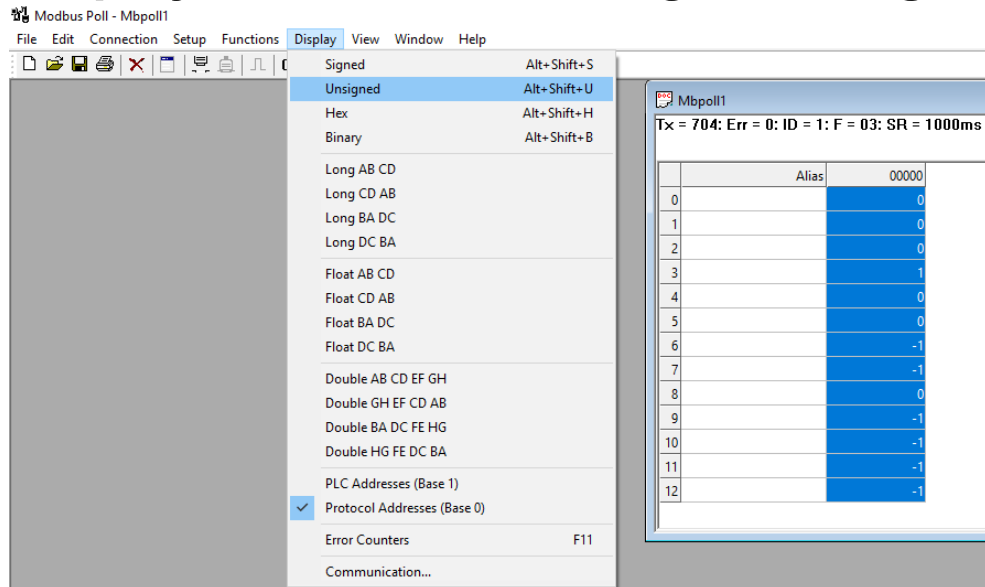
Address: 229

Quantity: 4

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6. Display all values as *Unsigned Integer*.



7. What is the value in register No. 4

Value:

Interpretation:

8. Read the serial number (Reg. No: 101, 102 ,103)

Value:

Interpretation:

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9. Read firmware version.

Value:

10. Read flow body temperature.

Value:

Interpretation:

11. Read relative humidity.

Value:

Interpretation:

12. Why does the EP015R6+BAC-HHM register No. 236 has a value of 65'535?

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13. Set the 6-way EPIV4 on *Position Control (Reg. No: 117)*.

Set Setpoint mode (Reg. No: 220) to Single setpoint.

Write the relative setpoint (Reg. No:1) of 10%.

Read out the *Active Sequence (Reg. No: 218)*.

Value:	Interpretation:

Write the relative setpoint of 90%.

Read out the *Active Sequence again*.

Value:	Interpretation:

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14. Connect 6-way EPIV4 to Sensor 5 field in MP Board.

Use potentiometer on MP Board to simulate analog sensor value and read sensor value at register no: 13.

Set also registers Setpoint Source and Sensor Type accordingly.

15. Read out Absolute Vnom in selected unit (Register no: 113/114).

See example in the Modbus Register Description on how to interpret 32-Bit values.

Value:

Interpretation:

16. Read out Error State (Register no: 157/158).

Value:

Interpretation:
